



The 21st Century Land Warrior, equipped with a GPS receiver, miniature camera, laser rangefinder, video and thermal weapons sights, and radio

non-lethal intoxicants to subdue the enemy.

The modern soldier on the ground still has to wade through mud, endure rough terrain and fight enemies, just the way armies fought with each other throughout the centuries. The difference is that the soldier of today has heightened senses, highly accurate and effective weapons, and an extensive array

of communication technologies. Body armour gets stronger and lighter; rifles, though still based on technology that's more than 50 years old, are now much more effective and equipped.

// **Let your plans be dark and impenetrable as night, and when you move, fall like a thunder-bolt** //

The art of war

As air power has developed, so have the armaments that give aircrafts their firepower. Missile systems offer a mixture of effects and ranges depending on the needs of the mission. Close-range sidewinders, long-range cruise missiles travelling several thousands of kilometres, laser guided bombs (LGB) that can be targeted from the air or by ground forces—the arsenal for air attack is ever increasing.

Laser guided bombs (LGB) such as the GBU-27 are specifically designed to be used with the F-117 Stealth Fighter. It is supplied with a 2000-pound bomb as armament, which holds 550 pounds of high explosive. A delayed action tail fuse allows the bomb to penetrate 6 feet of reinforced concrete before detonation. In association with the specialist F-117 aircraft, the GBU-27 can hit a square metre target from an altitude of 25,000 feet.

// **All warfare is based on deception** //

The art of war

Stealth. The ability to strike quickly and with surprise. Today's surface-to-air (SAM) missiles are equipped with tracking devices that are more sensitive—these missiles are able to chase down aircrafts from hundreds of miles away. So how does a warplane avoid detection? How do you make a huge plane disappear?

That was the challenge when research for the B-2 bomber began in 1981: to make an airplane invisible to the radar. The Pentagon originally wanted 132 planes. Only 21 were made. B-2 bombers are second-generation stealth bombers, made of carbon-graphite composites lighter than aluminium but stronger than steel. 136 separate computers onboard help the two crewmen handle the complex plane. With just one air refuelling, a B-2 can reach and strike accurately any target in the world. It has a capacity to carry up to 40,000 pound of weapons, including conventional and nuclear weapons, precision-guided munitions, gravity bombs and a range of maritime weapons. The B-2 can also carry the AGM-129 Advanced Cruise Missile,

## Information Warfare

Information warfare is a battle no longer done on a physical plane but a virtual one. Information technology is being developed as a weapon for 'logistics attack' as a means of disrupting the infrastructure on which an enemy's military apparatus depends. However, information warfare is a double-edged sword—the countries most capable of waging it are also the countries most susceptible to it.

Information warfare opens up new horizons of cost effectiveness for terrorists and enemy governments. Attacks on information



technology are unsettlingly easy to carry out. The means are relatively inexpensive, easy to smuggle, virtually untraceable, and completely deniable. This, coupled with the fact that the civilian networks are the most attractive to terrorists since they are the most vulnerable, makes 'Infowar' the perfect weapon in any terrorist's arsenal.

Future weapons of information warfare could include nano machines, electronic jamming,

high-energy radio frequency guns, and electromagnetic pulse bombs.